



Physics with Year Abroad, Literature Review: Particle Identification in Modern Cherenkov Imaging Detectors

Daniel Markhoff

School of Physics and Astronomy, University of Edinburgh, Edinburgh, UK and GSI Helmholtzzentrum für Schwerionenforschung GmbH, Darmstadt, Germany

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With major detector assemblies lining up for construction at high-energy facilities such as BNL, FAIR and KEK, technologies for particle identification are becoming ever more relevant. This review introduces the relevant theory behind the field of detector physics, motivates the technologies behind modern Cherenkov imaging detectors, and discusses the implementation of these systems into upcoming experiments. Emphasis is placed on particle classification techniques and the challenges presented by cutting-edge physics programmes. Persistent challenges include the continued development of photosensitive devices and deploying modern software and electronics to detector systems. We conclude by outlining promising future developments in the field.

Edinburgh Advisor: Moritz P Reiter

Placement Supervisor: Jochen Schwiening

mreiter@exseed.ed.ac.uk

j.schwieining@gsi.de

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I. INTRODUCTION

Modern high energy physics experiments rely on the performance of detector systems to accurately study collision events. Cherenkov imaging detectors will play a central role in the upcoming generation of experiments in quantum chromodynamics, gluon research, and high-energy neutrinos. Improving the capabilities of these imaging detectors is the work of thousands of physicists and engineers around the world.

This review aims to provide an introduction to the field of detectors making use of Cherenkov radiation for particle identification (PID). We will cover the basic principles of Cherenkov radiation, some design choices of Cherenkov imaging detectors, and briefly the latest advancements in technology. Cherenkov imaging detectors can be broadly divided into track-based PID systems, such as RICH (Ring Imaging Cherenkov) and DIRC (Detection of Internally Reflected Cherenkov) detectors used for PID at accelerators, and large-volume systems, such as large-volume water or ice Cherenkov detectors employed in neutrino and astroparticle physics.

We will begin with an overview of Cherenkov radiation in Sect II, and then introduce photomultipliers in Sect III. Following this, we will delve into track-based Cherenkov imaging detectors in Sect IV, where over the chapters we will discuss their designs and differences. Then, an astrophysics application to a topological Cherenkov tracking detector is introduced in Sect V. Finally, we will discuss some of the main points of open research in Sect VI.

II. CHERENKOV RADIATION

Cherenkov radiation is a phenomenon that occurs when a charged particle travels through a dielectric medium with refractive index $n(\lambda)$ at a speed greater than the phase velocity in that medium. This effect was first observed by Pavel Cherenkov in 1934¹ and later explained theoretically by Ilya Frank and Igor Tamm².

The angle of Cherenkov radiation θ_c relative to the particle track, for a particle of speed βc in a medium with refractive index n is

$$\cos \theta_c = \frac{1}{n\beta}. \quad (1)$$

From Maxwell's equations arises that electromagnetic propagation with wavelength λ will have its phase velocity modified by the medium it is travelling through. This leads to the idea of refractive index $n(\lambda) = c/v_p$, defined as the ratio between the speed of light in vacuum c and the phase velocity v_p in the medium. The condition for Cherenkov radiation to occur is then $v > c/n$.

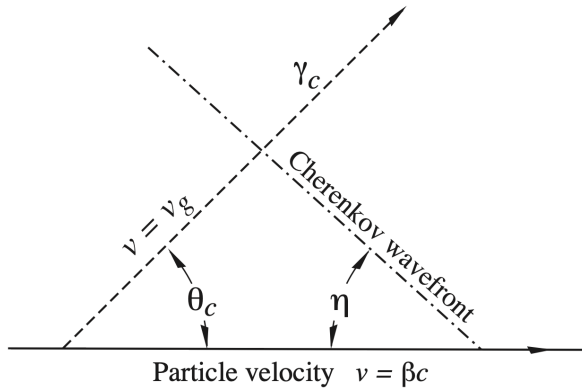


FIG. 1 **Cherenkov emission of photons and wavefront.** A particle travels through a medium with velocity v . Photons are emitted at the group velocity v_g and angle θ_c to the path of travel of the particle, giving the characteristic blue shock cone. In a dispersive medium, $\theta_c + \eta \neq 90^\circ$ generally³.

Let the photons have frequency ω and wavenumber $k = 2\pi/\lambda$. They propagate at the group velocity, $v_g = d\omega/dk$. The photons are concentrated in a thin conical shell as shown in Fig 1, with an opening half-angle η as

$$\cot \eta = \left[\tan \theta_c + \beta^2 \omega n(\omega) \frac{dn}{d\omega} \cot \theta_c \right] d\omega \quad (2)$$

where we are considering a small frequency range. Unless the medium is non-dispersive, where $dn/d\omega = 0$, generally $\theta_c + \eta \neq 90^\circ$, and the Cherenkov wavefront ‘sideslips’ along with the particle³.

The number of photons emitted per unit length is given by the Frank-Tamm formula

$$\frac{d^2 N}{dx d\lambda} = \frac{2\pi\alpha z^2}{\lambda^2} \left(1 - \frac{1}{\beta^2 n^2(\lambda)} \right) \quad (3)$$

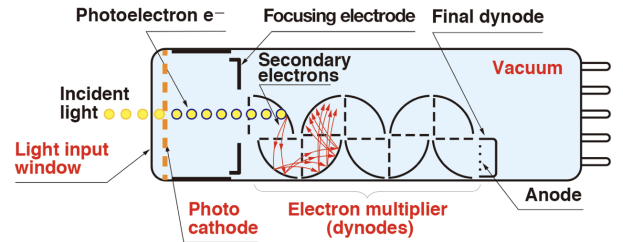
for a particle with charge ze and wavelength λ , moving through a medium with refractive index $n(\lambda)$, and where

α is the fine structure constant². For most uses, Equ 3 should be integrated over the region $\beta n(\lambda) > 1$. The number of photons N emitted per unit length is inversely proportional to the square of the wavelength, explaining the characteristic blue glow of Cherenkov radiation.

III. PHOTON DETECTION TECHNOLOGIES

III.A. PMTs

Photomultiplier tubes (PMTs) are photon sensitive detectors. A photon incident on a photocathode releases an electron through the photoelectric effect. These detectors then multiply the current from the electron by large amounts through emission of secondary electrons at the dynodes. This enables individual photons to be detected even when the incident flux of photons is low.



The best modern photocathodes used in Cherenkov imaging applications typically have a quantum efficiency (QE) of up to 30%, meaning that only 30% of incident photons satisfy the work function and produce a photoelectron⁴ that reaches the dynodes. The QE of a photocathode is wavelength dependent. For applications in imaging Cherenkov photons, a peak QE around the blue wavelengths is most desirable.

The gain of a PMT is the factor the photoelectron current is increased by at the anode. Modern technologies are able to achieve a gain of up to 10^8 , allowing individual photons to be easily detected. Optimising the dynode voltage, geometry and material are all factors that contribute to higher gain.

PMTs are simple and versatile devices that are still employed in many applications today, including Cherenkov imaging. However, due to the distance between dynodes, they are sensitive to magnetic fields. Since many detector systems at research facilities employ the use of solenoids for various experiments, they are often unfeasible. The solution is to significantly restrict the space between the dynodes.

III.B. MCP-PMTs

Micro-channel plate photomultipliers (MCP-PMTs) are an example of a photosensor that is resistant to magnetic fields. The micro-channel plates (MCPs) are thin wafers containing 5-20 μm diameter capillaries. Typically, two wafers are arranged such that the capillaries form a chevron pattern. Similarly to PMTs, electrons

produced via secondary emission with this MCP arrangement can achieve gains of 10^6 or more. This amplified electron signal is then recorded on an anode plate as an analogue voltage. The micro-channels of the MCP allow

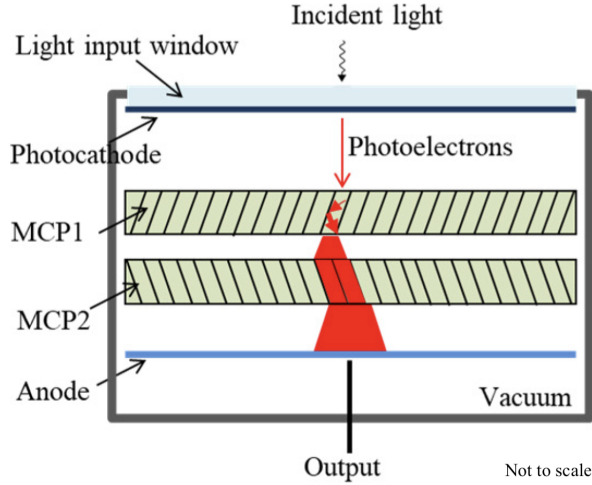


FIG. 2 **Cross section of a traditional MCP-PMT.** Due to the proximity of the anode and the photocathode, we are able to determine where the incident photon hit. MCP-PMTs are used in an incredible range of fields, from blood tests, to spectroscopy, laser radar⁵.

precise electron transit times towards the anode. They significantly reduce the path length difference between the electrons, leading to an extremely narrow signal temporal resolution. Modern devices achieve ≤ 50 ps single-photon timing resolution^{6–8}. A lower timing resolution permits more accurate event reconstruction and PID in Cherenkov imaging detectors.

The confinement of the electrons also helps to suppress any lateral motion due to e^-e^- repulsion or external magnetic fields. This improves the spatial resolution of the photosensor, as we are able to more precisely determine the location on the anode the signal originated from, indirectly improving both event reconstruction and PID.

However, while they are an incredible vacuum-based photosensor technology, the use of MCPs brings new challenges. For one, due to the large temporary depletion of local electrons inside the micro-channels after an incident photoelectron, the MCPs can quickly become saturated. This means that they struggle as the flux of incident photons picks up, often referred to as being rate limited.

MCP-PMTs are susceptible to dark counts. These are signals produced without an incident photoelectron. For MCP-PMTs, dark counts are dominated by thermionic emission at room temperature, field emission at high bias voltages, and positive-ion feedback during secondary emission avalanches. All of these processes produce charged particles, and they are able to damage the photocathode, with ion feedback causing the most issues. These processes can be minimised through cooling, vacuum conditions and shielding/coatings, yet they remain

an intrinsic lifetime limitation of the MCP-PMTs.

The photosensor introduced in the next section does not rely on photocathodes, vacuum sealing, and micro-channel pore etching as MCP-PMTs do. As such, they are cheaper to manufacture and pose a compelling alternative.

III.C. SiPMs

Silicon photomultipliers (SiPMs) have been gaining traction recently due to developments in materials, leading to a great ease of manufacturing. They are small compared to an MCP-PMT, due to their use of single photon avalanche diodes (SPADs), each containing an avalanche photodiode and a quenching resistor.

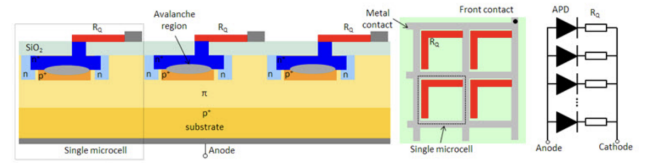


FIG. 3 **Cross section of SiPM.** SiPMs are externally biased such that each avalanche photodiode is at breakdown voltage, allowing for electron avalanches. The gain from an SiPM roughly matches that of a MCP-PMT at 10^5 – 10^6 amplification⁹.

SiPMs are overall cheaper per unit area than MCP-PMTs including all cryogenic and electrical supporting systems, and have a higher QE, however suffer from a high dark count rate and do not necessarily have similar hit rate capabilities, as the hit rate is mainly dependent on the density of SPADs. This has led to a surge of SiPMs being used in fields like positron emission photography (PET) where precise sub 50ps single-photon timing resolutions are not necessary. SiPMs also operate at far lower voltages than MCP-PMTs and are not affected by magnetic fields, adding to their dependability in medical settings.

The high dark count rate degrades PID in extremely precise systems, as noise causes statistical event reconstruction methods to struggle. This is mitigated with cryogenic cooling, however still remain around the MHz/cm² level, compared to the MCP-PMTs at only a few Hz/cm².

The dead time of SiPMs poses potential issues for systems which rely on high photon multiplicity for strong PID performance. The typical dead time of one SPAD is between 10–50ns¹⁰, so even though the microcells are much smaller than an MCP-PMT pixel, in the particularly photon-dense areas of the detector, it is possible lose some photons due to the dead time of the SiPM.

SiPMs provide an interesting alternative to MCP-PMTs, where their low cost and robust characteristics make them an lucrative option for many applications. Their performance in high-energy physics applications is certainly going to improve, and they are already a serious consideration for future LHCb RICH upgrades.

IV. TRACK-BASED DETECTORS

IV.A. PID via θ_c Measurement

PID in Cherenkov imaging detectors principally requires the measurement of the Cherenkov angle of a particle at a known momentum. Using the relativistic transformations for momentum and energy

$$p = \gamma m_0 v \quad (4)$$

$$E = \gamma m_0 c^2 \quad (5)$$

we can rewrite the Cherenkov angle from Equ 1 as

$$\cos \theta_c = \frac{\sqrt{p^2 + m^2 c^2}}{np}. \quad (6)$$

At a fixed momentum, the Cherenkov angle provides separation between particle hypotheses with different masses. This equation can be used to plot the analytical curves for the experimental data in Fig 4.

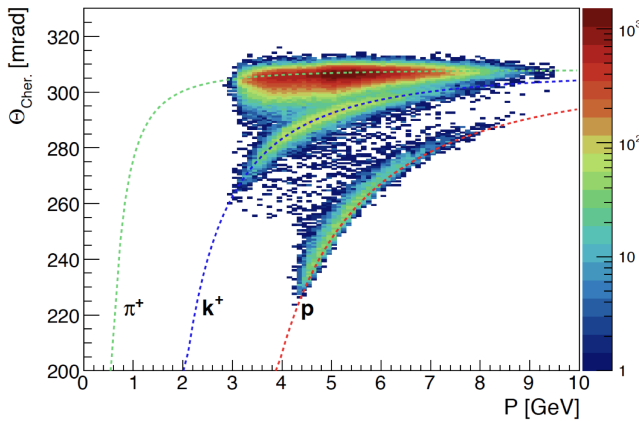


FIG. 4 **Cherenkov angle response of the CLAS12 detector at JLab¹¹**. There are distinct curves for different particle species. The analytical form of these curves can be determined using Equ 6.

In modern detector arrays at collider setups, forward tracking detectors provide momentum measurements. Then, Cherenkov track-based PID detectors (such as RICHs in Sect IV.B) provide measurement of the Cherenkov angle through analysis of the Cherenkov photon hits incident on a photodetector array. Large-volume detectors (such as Super-K in Sect ??) map the entire Cherenkov cone of photons using photodetectors over a huge area, and can obtain momentum estimates from the light intensity.

IV.B. RICH

Ring Imaging Cherenkov Detectors (RICH) are a class of Cherenkov detectors that use a radiator at a known distance from a photodetector array to measure the Cherenkov angle of incident particles. Shown in Fig 5 is a simplified Geant4 simulation of a RICH detector.

The Cherenkov photons are emitted in a cone around the particle track. Depending on physics goals, the detec-

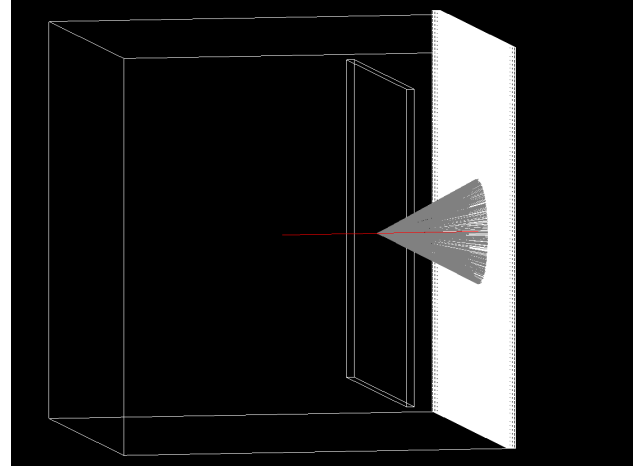


FIG. 5 **Geant4 simulation of the simplified RICH detector**. The particle track (red) is incident on some material with a refractive index n , called the radiator. In this case, an aerogel radiator is simulated. The resulting Cherenkov photons (gray) emitted in a cone are incident on a square array of sensitive photodetectors.

tor should not disturb the particle track for detectors further down the beamline. This motivates the ring-shape of some RICH photodetector arrays. The outer radius of the ring is often encased in mirrors or lenses to trap Cherenkov photons and guide them towards the photodetector array.

IV.B.1. Radiators

Different RICH experiments use different radiator materials. Common choices include aerogel such as in the Belle II ARICH¹², gaseous radiators (noble gases, halogenoalkanes) such as in the old LHCb RICH¹³ and (although in mostly retired detectors) liquids such as in the barrel CRID at SLAC¹⁴.

Consider the Cherenkov angle condition from Eq 1. The radiator refractive index n is the key parameter that determines the choice of radiator, depending on expected particle momenta. Higher refractive indices are necessary for lower momentum particles, which is why some accelerator systems use liquid radiators with inherently higher refractive indices.

The refractive index is also wavelength dependent, which is a source of chromatic dispersion. This is an important consideration for radiator materials, as chromatic dispersion can degrade angular resolution. The refractive index also affects the number of Cherenkov photons produced, as per Eq 3. There are also tricks to play with the radiator design to improve photon yield, such as layering aerogel with different refractive indices without performance cost¹⁵. Naturally though, complexity comes at a monetary cost.

IV.B.2. Optics (CLAS12 RICH)

Optical elements are often used in RICH detectors to increase photon yield. Lenses and mirrors can be used to focus Cherenkov photons onto smaller photodetector areas, reducing cost and size. Other optical tricks such as the aforementioned layered radiator (also called proximity focusing) can be used to tighten the spread of the photon cone, allowing lower momentum charged particles to be measured.

As an applied example, the RICH at the forward section of the CLAS12 detector at JLab makes use of spherical mirrors to actively guide photons towards a small area of photodetectors. This both improves the photon yield and the angular acceptance of the detector. More photons means better angular resolution and therefore improved event reconstruction.

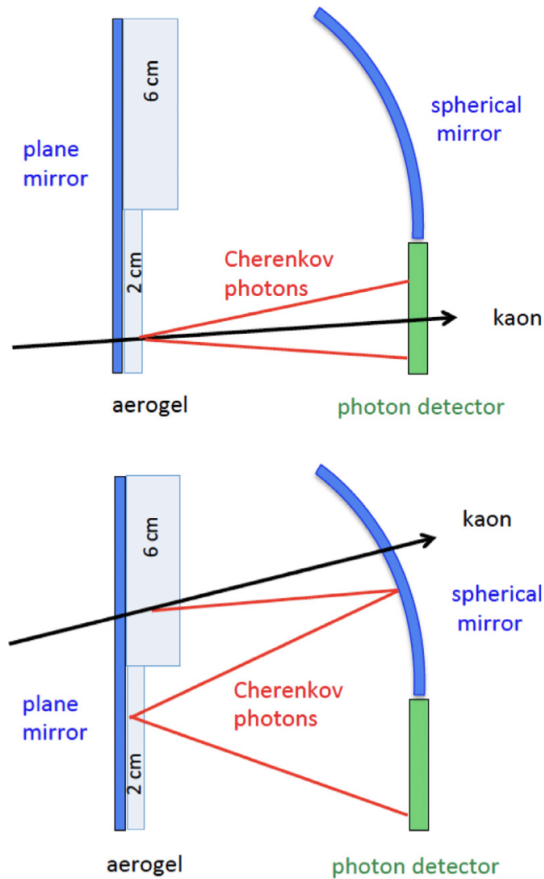


FIG. 6 **Examples of CLAS12 RICH active focusing.** The spherical mirrors near the edge of the vessel mean that Cherenkov photons can be detected from steeper angles than would be traditionally possible.¹¹

Properly designed active optical elements in Cherenkov imaging detectors provide the possibility to reduce the total cost of the photodetector array by guiding photons to a smaller area, while simultaneously improving the photon yield and therefore the statistics of the measured Cherenkov angle.

IV.C. Statistical Reconstruction & PID Frameworks

In modern Cherenkov detectors, event reconstruction and particle identification are often performed within a unified statistical framework. In the following, we distinguish conceptually between the reconstruction of event observables and the subsequent PID decision, while noting that many algorithms perform both tasks simultaneously.

The problem of event reconstruction in modern Cherenkov detectors is to determine Cherenkov photon paths inside the detector based on the observable data obtained from the photon sensors. The most common method of doing this currently is through reverse ray-tracing. An algorithm iterates through the possible paths a photon can take through the detector, and evaluates a Gaussian likelihood under the hypothesis of each particle type¹⁶. This likelihood is built from the spatial and temporal residuals between the detected photon hits and the expected photon hits from the ray-tracing. For example, a Gaussian spatial likelihood can be expressed as:

$$\mathcal{L}_h = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \langle x_{i,h} \rangle)^2}{2\sigma^2}\right) \quad (7)$$

where x_i is the detected position of the i -th photon, $\langle x_{i,h} \rangle$ is the expected position of the i -th photon under hypothesis h , and σ is the spatial resolution of the detector system. The hypothesis h can be for example an incident pion or a kaon. Note that many algorithms choose to use non-Gaussian likelihoods. By calculating the likelihood for each hypothesis, we can then form a log-likelihood ratio to determine which hypothesis is more probable:

$$\Delta \log \mathcal{L} = \log \mathcal{L}_{\text{kaon}} - \log \mathcal{L}_{\text{pion}} \quad (8)$$

This can be minimised across all detector hits to find the most probable particle type using various numerical methods. This method therefore implicitly reconstructs the observables, and then uses these observables with the end goal of identifying the incident particle. These implicitly reconstructed observables can be used as metrics for detector performance.

Other likelihood-based methods include time-imaging¹⁷, where precise photon timing information is also used to build probability density functions (PDFs) as a supplement to spatial PDFs for each particle hypothesis. The detected photon times are then compared to these PDFs to form likelihoods. Many modern algorithms use a combination of spatial, temporal, angular information from other sub-detectors, and even photon count information to build more robust likelihood functions.

Likelihood-based methods are among the most powerful and widely used analysis techniques in countless experiments. Their practical application, however, often requires detailed construction and evaluation of high-dimensional likelihoods, leading to significant computational overhead in complex detectors.

Machine learning (ML) approaches offer a promising alternative by replacing PDF construction with data-driven inference. Neural networks can be trained to learn the detector response to different particle hypotheses directly, thereby implicitly capturing the information normally encoded in carefully constructed PDFs. This has already seen encouraging results on Geant4-simulated data for the Hyper-Kaminokande experiment¹⁸ and for the PANDA Barrel DIRC¹⁹.

IV.D. DIRC

Devices that detect internally reflected Cherenkov light are a class of Cherenkov imaging detectors, called DIRCs for short. They use solid transparent bars or plates as Cherenkov radiators, made typically from fused silica quartz, to both generate and guide Cherenkov photons to the photodetectors. Fig 7 shows a detailed rendering of the PANDA barrel detector and the components making up one sector²⁰.

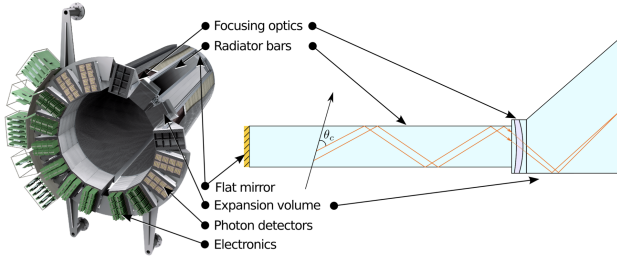


FIG. 7 **The PANDA Barrel DIRC along with an individual bar cross section.** 16 individual sectors are laid in a barrel-like arrangement, forming the DIRC detector. Photons are internally reflected through the quartz bars. They are focused into an expansion volume, and then incident onto a photodetector array.

The basic principle of DIRC operation is equivalent to that of RICH detectors; charged particles incident on a medium with refractive index $n > 1$ emit Cherenkov photons in a cone, and we try to measure the principle angle of this cone. However, in a DIRC detector the complexity arises from the optics of trapping and guiding the Cherenkov photons to a small array of photodetectors. This necessitates the use of statistical methods for PID, rather than direct measurement as in RICH detectors.

IV.D.1. Photon Transport via Internal Reflection

Similarly to photon transport in optical fibres, the Cherenkov photons produced in the quartz bars of a DIRC are trapped via total internal reflection as shown in Fig 7. The photons propagate to the end of the bar, where they are focused into an expansion volume, and then onto the photodetector array. The DIRC relies on photons internally reflecting hundreds of times with minimal loss, all while preserving the Cherenkov angle. The

fused silica bars provide extremely smooth surfaces to facilitate this, with surface roughness generally possible on the order of 3-10 angstroms²¹.

The quartz bars are also UV transparent, allowing Cherenkov photons in the near-UV range to be detected, increasing the photon yield of the detector. Fused silica quartz is also known to be radiation hard, which will allow the detector to operate in the high radiation environments of modern accelerators without significant degradation over time²². The fused silica bars also have a good refractive index of around 1.47 in the visible spectrum, giving large Cherenkov angles, without pushing too many photons into the undetectable UV range.

IV.D.2. Chromatic Effects & Timing Resolution

Since Cherenkov photons are produced across a range of wavelengths, chromatic dispersion is unavoidable. Since the quartz bars are often meters long, blue photons often arrive hundreds of picoseconds later than red photons. This is due to the wavelength dependence of the refractive index, which is around 1.50 for blue light and 1.47 for red light in quartz. However, dispersion can be corrected for if the photon arrival time is measured with sufficient precision, making it a crucial aspect of DIRC design.

Modern MCP-PMTs are able to achieve single-photon timing resolutions $\leq 50\text{ps}$ ⁶⁻⁸. Improvements in readout electronics and signal processing are also pushing timing precisions lower²³. These advancements allow photon timing to become a far more powerful observable; detected photons provide accurate x, y, t rather than just x, y . We can bake this information into our event reconstruction algorithms, simulating expected photon arrival times for bluer and redder photons separately, massively mitigating the effects of chromatic dispersion in the fused silica.

These sorts of chromatic corrections via timing information lead to far more sophisticated event reconstruction than RICH systems. As discussed in Section IV.C, statistical methods such as maximum likelihood estimation are extremely powerful. DIRC systems offer a unique double-edged sword; complex photon paths, internal reflections, and chromatic dispersion make analytical solutions near impossible, but the extremely detailed information available from precise timing measurements allow for robust statistical models such as time-imaging¹⁷ and ML to shine.

IV.D.3. Reconstruction Methods & PID

The aforementioned time-imaging reconstruction method builds up a PDF from expected photon arrival times for all detector array channels and particle types. Then, events to be classified can be compared to these PDFs on a maximum likelihood basis. The algorithm can be optimised using a lookup table of possible paths a photon could take to a specific photosensor pixel,

eliminating a few unnecessary computations. This allows more flexibility in focusing optics within the detector geometry, as possible paths can be analytically determined and placed in the lookup table. This improves photon yield and therefore PID performance¹⁷. However, from a software standpoint, the algorithm is unfriendly to parallelise due to heavy use of conditional logic. Parallel hardware prefers to perform the same operation on many different data points. This algorithm demands different operations depending on detector channel, timing, and hypothesis. As such, computation time becomes a problem for large datasets.

IV.D.4. Machine Learning

ML on the other hand is significantly easier to parallelise. Raw detector data is fed to the network on a per-event level and a sequence of fixed operations (mostly matrix multiplications) are performed. These ML models are easy to distribute onto modern hardware. Since their fixed operation processing speed is predictable per event, as it is correlated to model size, data processing is far less likely to bottleneck; we are able to process batches of events at a time and can reliably expect all events to finish processing at around the same time. This is in contrast to variable latency such as in time-imaging reconstruction, where often batches are waiting on one particularly difficult event to finish processing before starting the next batch. These advantages make ML scaleable in a way that maximum likelihood algorithms are not.

Many different neural network architectures are being researched. This refers to the order and manner of the sequence of fixed operations that are done by the network. Dense networks consist of purely matrix multiplications, yet are seeing promising results¹⁹. More complicated networks such as normalising flows are closer to previous PDF construction methods, but offer exact likelihood evaluation without iterative maximisation, combining the known strong performance of the PDF-based methods with the compute advantages of ML methods²⁴.

While such approaches have demonstrated significant gains in PID performance, their impact must be understood within the broader context of detector design, where the available information is fundamentally constrained by the underlying hardware and physics requirements.

IV.E. Mechanics & Design of Modern Detectors

Modern detector systems are large, complicated devices that enjoy premium real estate. Integrating a Cherenkov detector (or any detector, for that matter) into such a system presents a unique set of mechanical and design challenges. Mechanical tolerances and alignment are crucial to maintaining accuracy. Optical elements must be precisely aligned and consistently calibrated throughout the detector lifetime, requiring robust mechanical support and space-hungry laser calibration

systems. Careful thermal and radiation management are also necessary to ensure the longevity of photosensitive components. Consideration of the surrounding magnetic field from the solenoid used by the TOF detectors is also necessary, especially for PMTs which are magneto-sensitive.

Ensuring that the detector can be serviced and maintained is also a key consideration, especially for large scale experiments that may run for decades. Some experiments are built on rails to allow the inner detectors (RICH, TOF, trackers etc.) to be slid out for maintenance. These considerations are imperative to the success of a detector.

RICH detectors require a large volume radially to generate and image the cone of Cherenkov photons. DIRC detectors on the other hand, are far more radially compact, offering an advantage in certain facilities. However, the detection of natural events is not possible with either of these systems. If one wanted to study the particles incident on Earth from outer-space, a different system entirely is needed.

V. TOPOLOGICAL DETECTORS FOR NEUTRINO ASTROPHYSICS

Topological Cherenkov detectors function more as particle trackers than devices optimised for particle-species discrimination. They image Cherenkov photons across large distances of particle tracks, measuring interaction topology, particle direction, and energy, rather than single-track observables such as the Cherenkov angle θ_c . Since neutrino interactions are rare and large in scale, the detector and radiator material must span an enormous volume to obtain a sufficient flux. The simple solution to this is to use large natural volumes of material such as water or ice. This presents a unique set of engineering challenges, such as waterproofing the detectors and synchronising timing across large distances. It is also very difficult to separate the weakly interacting neutrino events from other background. Different collaborations have taken unique approaches to these problems depending on their physics programmes.

V.A. Sources of Neutrinos

There are many astrophysical sources of neutrinos of current interest, such as active galactic nuclei, neutron stars, gamma-ray bursts, and supernovae, ranging from MeV to PeV energies. These neutrinos are predominantly products of pion, muon and neutron decays.

$$\begin{aligned}\pi^+ &\rightarrow \mu^+ + \nu_\mu \\ \mu^+ &\rightarrow e^+ + \nu_e + \bar{\nu}_\mu \\ n &\rightarrow p + e^- + \bar{\nu}_e\end{aligned}$$

The relative contribution of these channels depends on the production environment and neutrino energy. There are also ultra-high energy sources of neutrinos, ranging from PeV to EeV energies. Protons near black

holes are accelerated to incredible velocities where they are able to interact with the cosmic microwave background. Then, the produced pions decay into extremely fast-travelling neutrinos. This process is known as the Greisen-Zatsepin-Kuzmin (GZK) process, as depicted in Fig 8.

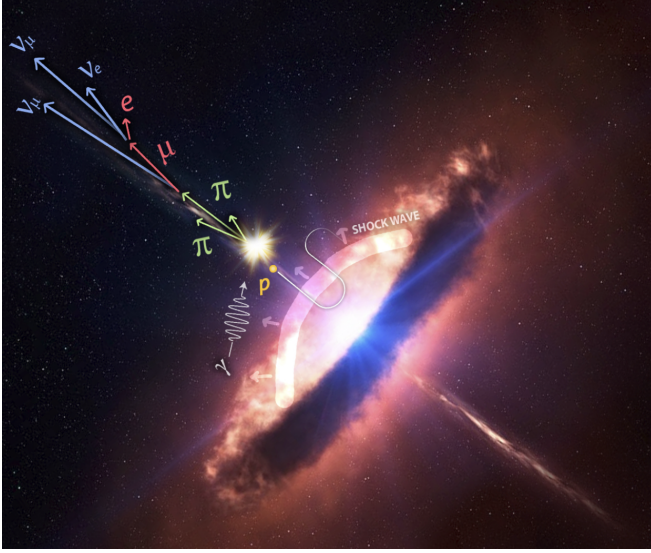


FIG. 8 **Cosmic neutrinos formed through the GZK process.** Protons accelerated by black holes are able to interact with the cosmic background, decaying through intermediary steps into various high energy neutrinos²⁵.

Detectors investigating these processes must use kiloton-gigaton volumes of radiator material to track these neutrinos due to their high energies and minimal cross-section. As such, the material used must be cost effective and easy to handle.

V.B. Water Cherenkov Detectors (Super-K)

Water Cherenkov Detectors (WCDs) use large volumes of ultra-pure water as the radiator material. The Super-Kamiokande (Super-K) experiment in Japan is a famous example of a WCD, using 50,000 tons of water surrounded by over 11,000 PMTs, buried under a kilometre of mountain to shield from natural radiation. The detector is used to study neutrinos at MeV-GeV energies from various sources, including the sun, atmosphere, supernovae. The detector has investigated neutrino-specific phenomena²⁶, which before Super-K were largely unexplored.

Super-K is a precision detector for low energy neutrino physics. If we wish to study higher energy phenomena, such as neutrinos generated by the GZK process, we need to significantly increase the scale of the detector. This means that using pure water is unfeasible, and we must search for natural features on the Earth to act as a Cherenkov radiator.

V.C. Ice Cherenkov Detectors (IceCube)

The IceCube Neutrino Observatory is a cubic kilometre scale Cherenkov detector buried deep in the natural Antarctic ice sheet²⁷. It uses over 5,000 photodetector modules arranged on 86 strings, deployed between 1.5 and 2.5 km deep. The intra-string separation varies from 41 - 105m. The detector is designed to study ultra-high energy neutrinos. These neutrinos are filtered from low energy neutrinos and other background by using the entire Earth as a shield; up-going events are strongly enriched in high-energy neutrino interactions, as atmospheric muons are absorbed by the Earth.

Fig 9 shows such a cosmic event from IceCube, where a high energy neutrino has interacted with the ice to produce muons and subsequent Cherenkov radiation. The thickness of the strings represents photon intensity, while the colours represent timing information, red being at the beginning of an event and green/blue being later in time.

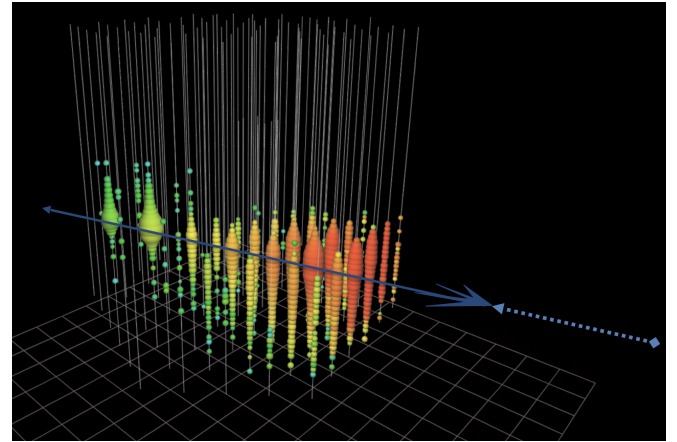


FIG. 9 **IceCube event display.** Multiple charged particle tracks can be distinguished. Maximum likelihood methods can be used to reconstruct the muon tracks in the ice²⁵.

Cherenkov detectors have proven to be invaluable tools in the current study of physics, but they also have a promising future. We will now discuss the main points of open research in Cherenkov-based detector physics.

VI. CURRENT CHALLENGES & LIMITATIONS

Much of modern detector research is placed into improving photosensor performance in high-rate collider environments. Along with rapidly developing SiPMs, improvements are also being made to MCP-PMTs, both towards better spatial and temporal resolution. A process known as atomic layer deposition (ALD) coats the MCPs with resistive layers to precisely tune the performance²⁸; this has been shown to significantly improve sensor ageing²⁹. Research into Large Area Picosecond Photodetectors (LAPPDs) are attempting to reduce the cost per area of traditional MCP-PMTs while matching their performance characteristics. LAPPDs themselves

have been shown to have good spatial and temporal resolution³⁰, and further characterisations of HRPPDs, a high-rate upgrade for LAPPDs, are ongoing^{31,32}.

On the software side developments being researched are as described in Sect IV.D.3 for DIRC systems. ML has the potential to offer fast particle classification in all Cherenkov imaging systems, either alone or in conjunction with other PID methods to decrease compute time. Effort is being focused into reducing model size, whilst retaining PID performance. This makes the models easier to deploy on the beamline and improves classification speed.

On the electronics side, developments are increasingly driven by the demand for precise timing and high-rate readout to fully exploit modern photosensors. Fast sampling ASICs³³ and high-bandwidth digitisation schemes³⁴ are required to preserve the picosecond scale timing resolution of MCP-based detectors. In high-rate environments, advances in front-end electronics and data acquisition systems aim to reduce dead time and manage large data volumes while maintaining low noise and power efficient performance. Close integration of sensors with dedicated readout electronics is therefore an important area of research.

Finally, reducing the cost of detector systems will always be of great interest. The ease of manufacturing for SiPMs offer the potential to significantly reduce the price tag on detector arrays. Reuse of retired detector parts is also a potential cost saving measure; the recent transport of legacy BaBar boxes for use in the hpDIRC at Brookhaven is one example of successful reuse³⁵.

VII. CONCLUSION

This review introduced the field of Cherenkov imaging detectors, and covered modern developments towards precision PID in upcoming nuclear and high-energy physics experiments. The detectors of today will be able to further our understanding of quantum chromodynamics, gluons in hadron events, and elusive high-energy neutrinos.

Highlights from the developments into the detectors include the push to improve spatial and timing resolutions of photosensors, with many difference devices reaching impressive single-photon time resolutions. The lifetime and radiation resistance of the photosensors has also been developed, with novel fabrication techniques such as ALD. Furthermore, ML for particle classification is one of the fastest moving areas of the field. They have the potential to reduce computation time compared to other statistical PID methods depending on the model size, and have been shown to match the accuracy of the old methods with large detailed models.

There is no shortage of improvements to be made to these detectors. As physics programmes demand higher luminosities and higher energies, the technology must adapt. There are clear limitations in the high-rate ca-

pability of photosensors and their DAQ systems. The size of data per second expected to be generated by near future detectors is immense. The cost of scientific programmes is ever more scrutinised, and so refinements towards the manufacturing of detector components are an area of particular focus.

Nevertheless, it is an exciting time for detector physicists; the FAIR international accelerator facility and the HL LHC upgrades are both due to complete construction before 2035. The Electron-Ion Collider at BNL is due to begin operations after 2035. Hyper-K, the successor to Super-K in Japan, is a neutrino detector that began civil engineering works and excavation 2 years ago.

Cherenkov imaging detectors have and will continue to be instrumental in high-energy physics research. They are a tried and true method of identifying charged particles. In the modern era, the detectors are beginning to move beyond auxiliary devices for separate mass hypotheses. Rather, they are playing vital roles in entire event reconstruction, allowing physicists to better study the interactions of our universe.

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